

REJUVENATION STRATEGY OF DYING SPRINGS AND NAULAS IN DHARGAD RAINFED RIVER WATERSHED, JAIGAN VALLEY, DISTRICT BAGESHWAR, KUMAUN HIMALAYA, (UTTARAKHAND)

PHASE-I

REJUVENATION OF SPRINGS AND NAULAS OF JATHA AND PASS VILLAGES IN DHARGAD WATERSHED

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EXECUTIVE SUMMARY

The Himalayan region, often referred to as the "Water Tower of Asia," is experiencing a severe crisis both in glacial-fed and non-glacial rainfed river watersheds. Rainfed rivers in Uttarakhand play a vital role in supporting agriculture, industry, and human consumption, having over 353 significant rainfed rivers of fourth order and above spanning 5,796 km. Within the watersheds of these rainfed rivers, about 70% of villages and more than 80% of urban centers of Uttarakhand are located.

Due to anthropogenically and technogenically accelerated climate change impacts, traditional water sources such as natural springs and naulas are steadily transforming from their perennial nature to ephemeral nature, causing major rivers to dwindle rapidly due to decrease in base flow during summer seasons. Failing to take planned scientific action for groundwater augmentation through massive rainwater harvesting mechanical and biological measures in the recharge areas will lead to dire consequences, including severe drinking water scarcity, biodiversity loss, and exacerbated climate change impacts.

This report formulates a scientific integrated strategy for rejuvenation of springs, streams, and rivers using the Dhargad River watershed in Bageshwar district as a natural laboratory.

Fundamental Objectives:

1. Identify and prioritize dead and dying naulas, springs, and streams in the watershed.
2. Assess the status of water sources and delineate potential hydro-geomorphic recharge zones.
3. Develop a GIScience-based Decision Support System (DSS) for rejuvenation planning.
4. Implement and monitor micro-plans in active association with local inhabitants and Gram Panchayats.
5. Mitigate the adverse impacts of climate change on hydrology and mountain livelihoods.

1.0 INTRODUCTION & PROBLEM STATEMENT

The Dhargad River watershed is a tributary watershed of the Jaigan River which finally joins the Saryu River at Lweta village (elevation 716m, 29°42'13.551"N Lat and 79°53'40.896"E Long) in the Bageshwar district of Kumaun Himalaya. It represents a classic case of rainfed Himalayan catchments undergoing acute hydrological degradation.

Consequences of Inaction:

- **Water Scarcity:** Severe shortages of drinking water during 2-3 peak summer months for human settlements and livestock.
- **Ecosystem Degradation:** Drying of aquatic biodiversity, loss of soil moisture, and abandonment of terraced agriculture.
- **Climate Vulnerability:** Increased cloudburst runoff velocities causing rill and gully erosion without groundwater infiltration.

3.0 ABOUT THE DHARGAD RIVER WATERSHED

The entire Dhargad watershed encompasses an area of 2.32 km² lying between 29°43'38.117"N to 29°44'53.908"N Latitude and 79°43'39.756"E to 79°44'47.547"E Longitude. The altitude varies from 1,295 m to 1,492 m MSL. It is composed of the Raut Gara Formation (Damta Group) characterized by quartzites interbedded with silky phyllites and slates.

Phase-I focuses on the experimental recharge zone of Village Jatha and Pass covering 0.17 km² (17 hectares).

TABLE-1: DETAILS OF SPRINGS AND NAULAS OF THE STUDY AREA

Source: Field Survey by Jagdish Chandra & Team (AMAN NGO)

S.N.	Water Source	Type	Elevation	Families (Pop.)	Past (10 yr)	Present Status
1	Dhargad Spring	Spring	1324 m	80 (350)	Working	Working (0.34 L/s)
2	Bhaidyon Spring	Spring	1441 m	62 (150)	Working	Working
3	Syari Naula	Naula	1362 m	12 (25)	Working	Working
4	Mathi Naula	Naula	1397 m	5 (11)	Working	Completely Damaged
5	Nikai Naula	Naula	1306 m	Nil (Dry)	Working	Completely Damaged
6	Bhairav Naula	Naula	1397 m	30 (110)	Damaged	Completely Damaged
7	Thaman Naula	Naula	1449 m	30 (110)	Working	Damaged (Summer Only)

TABLE-2: DISTRIBUTION OF AREA UNDER DIFFERENT RELIEF ZONES

Relief Zone	Area (km²)	Percentage (%)	Hydrological Classification
> 1400 m	0.03 km²	17.65%	High Relief (Recharge Ridge)
1350 - 1400 m	0.05 km²	29.41%	Medium Relief (Terraced Interventions)
< 1350 m	0.09 km²	52.94%	Low Relief (Discharge & Valley Floor)
Total Area	0.17 km²	100.00%	Phase-I Experimental Zone

TABLE-3: ESTIMATED NUMBERS OF MECHANICAL TREATMENT MEASURES*Engineered for Phase-I Recharge Area (0.17 km²)*

Measure Name	Forest Land	Barren Land	Aband. Agri	Total Planned
Infiltration Holes	90,000	160,000	150,000	400,000 units
Infiltration Trenches	291	2,050	-	2,341 units
Bio-Plugs	-	-	20 across 2 rills	20 units
Rill/Gully Plugs	-	-	5 across channels	5 units
Contour Bunds	-	-	5,000 running meters	5,000 meters
Terrace Ponds	-	10 ponds	20 ponds	30 ponds
Check Dams	-	-	2 near rill mouths	2 units

SPECIFICATIONS & STANDARDS OF MEASURES:

- Infiltration Holes: 10-15 cm depth, 5 cm diameter, density 5 holes/m² constructed with steel bars in barren/abandoned fields to catch high-intensity rain drops.
- Infiltration Trenches: Staggered cuts of 1.0m × 0.5m × 0.3m spaced 5-10m apart on convex hill slopes to break sheetwash runoff and force deep percolation into bedrock fissures.
- Bio-Plugs & Gully Plugs: Constructed with pine needles, local logs, and stones across first-order ephemeral rills to arrest velocity and trap sediment.
- Terrace Ponds: 10-20 m² area, 1-2 m depth shallow retention ponds with earthen embankments constructed in association with villagers.

BIOLOGICAL ECO-RESTORATION & COMMUNITY RESOLUTIONS

Biological Treatments:

- Afforestation: Planting native broad-leaved water-harvesting species across 0.032 km² barren lands, specifically Banj Oak (*Quercus leucotrichophora*), Uthis/Alder (*Alnus nepalensis*), Pipal, and Buransh (*Rhododendron*).
- Vegetative Mulching: Planting Napier and Vetiver soil-binding perennial grasses along trench banks.
- Economic Horticulture & Floriculture: Utilizing 0.132 km² of abandoned terraces for community nurseries, fruit trees, and commercial floriculture.

Annexure-I: Gram Panchayat Resolutions (November 29, 2025):

A formal workshop was organized by AMAN NGO in Village Jatha and Village Kafaligair / Dharalkhunga. Led by Prof. J.S. Rawat, Jagdish Chandra, and Neelima Bhatt, the village meetings unanimously resolved to initiate immediate groundwater augmentation focusing on Ward No. 6 and Ward No. 7 in Jatha village.